Foundations of Symmetry in Quantum Theory

Core Concept

A symmetry is a transformation of quantum states that preserves all physical observables, implying indistinguishable experimental outcomes.

Key Definitions

- **Symmetry**: Transformation preserving observables.
- **Hilbert Space H**: Space of quantum states.
- **Unitary**: Operator U preserving inner products ($U^*U =$

Mathematical Structure

Symmetry: $\phi \to U\phi$ where U is unitary or antiunitary

Wigner's Theorem. U = Unitary or Antiunitary

Transition Probability: $|\langle \phi, U\varphi \rangle|^2 = |\langle \phi, \varphi \rangle|^2$

Concept Map / Diagram

Physical Interpretation

- Observables unchanged by symmetry
- Unitary: continuous symmetries (rotations, time evolution)
- Antiunitary: discrete symmetries (time reversal)

Connections

- Conservation Laws
- Lie groups, Lie algebras
- Noether's Theorem
- Poincare Group
- Gauge Symmetries

Final Insight

Symmetry defines what is physically unboservable,

Symmetries represent redundancies in description that correspond to fundamental conservation laws.